

The Syllabus for Basic Academic Abilities in the EJU
(To be applied to the questions of the 2026 EJU 1st Session)

〈 Syllabus for Science 〉

[Purpose of the Examination]

The purpose of this examination is to test whether international students have the basic academic ability in science necessary for studying at universities or other higher educational institutions in Japan.

[Classification of the Examination]

The examination consists of three subjects: physics, chemistry, and biology. Examinees are required to select two of these subjects.

[Scope of Questions]

The scope of questions is as follows. What is taught in elementary and junior high schools in Japan is regarded as already learned and therefore to be included in the scope of the EJU. For each subject, the contents are grouped and presented by topic titles or key scientific terms.

Chemistry

The scope of questions will follow the scope of “Basic Chemistry” and “Advanced Chemistry” of the Course of Study for high schools in Japan.

I Structure of Matter

1. Composition of matter and constituent elements
 - (1) Pure substances and mixtures
Pure substances, mixtures, separation and purification of substances
 - (2) Matter and its composition
Elements, simple substances, compounds, allotropes, detection of elements (flame reactions, precipitation reactions)
 - (3) States of matter
Three states of matter (gas, liquid, and solid), changes of state
2. Constituent particles of matter
 - (1) Structure of the atom
Electrons, protons, neutrons, atomic number, mass number, isotopes
 - (2) Electron configuration and periodic law
Electron shell, properties of atoms, periodic table, outermost electrons, valence electrons
3. Substances and chemical bonds
 - (1) Ionic bonds
Ionic bonding, ionic crystals, ionization energy, electron affinity, electrolytes
 - (2) Metallic bonds
Metallic bonding, free electrons, metallic crystals, malleability and ductility

- (3) Covalent bonds
 - Covalent bonding, coordinate bonding, covalent crystals, molecular crystals, polar nature of bonds, electronegativity
- (4) Intermolecular force
 - Van der Waals force, hydrogen bonding
- (5) Chemical bonds and properties of substances
 - Melting point, boiling point, density, solubility, electric conductivity, thermal conductivity
- 4. Quantitative treatment of substances and chemical formula
 - (1) Amount of substance
 - Atomic weight, molecular weight, formula weight, amount of substance, molar concentration, mass percent concentration, molality
 - (2) Chemical formulas
 - Molecular formula, Lewis structures, structural formula, compositional formula (empirical formula)

II State and Change of Substances

- 1. Change of substances
 - (1) Chemical reaction
 - Chemical equation, stoichiometry
 - (2) Acids and bases
 - Definition and strength of acids and bases, hydrogen ion concentration, pH, neutralization, acid-base titration, salt
 - (3) Oxidation and reduction
 - Definition of oxidation and reduction, oxidation number, ionization tendency of metal, oxidizing and reducing agents, redox titration
- 2. State and equilibrium of substances
 - (1) Change of state
 - Thermal motion of molecules and three states of substances, energy distribution of gaseous molecules, absolute temperature, boiling point, melting point, heat of fusion, heat of vaporization, vapor pressure, phase diagram
 - (2) Properties of gases
 - State equation of ideal gas, mixed gas, law of partial pressure, real gas and ideal gas
 - (3) Solubility equilibria
 - Dilute solution, saturated solution, solubility product, supersaturation, solubility of solids, solubility of gases, Henry's law
 - (4) Properties of solutions
 - Depression of vapor pressure, elevation of boiling point, depression of freezing point, supercooling, osmotic pressure, colloidal solution, Tyndall effect, Brownian motion, dialysis, electrophoresis
 - (5) Structure of solids

Crystals, amorphous solids, unit cell, metallic crystals (body-centered cubic lattice, face-centered cubic lattice (cubic close-packed structure), hexagonal close-packed structure), ionic crystals, covalent crystals, molecular crystals

3. Chemical reaction and its equilibrium

(1) Chemical reaction and energy

Heat and light in chemical reactions, exothermic reaction, endothermic reaction, bond energy, enthalpy, reaction enthalpy, enthalpy of formation, Hess's law

(2) Electric cell

Electrical energy and chemical energy, Daniell cell, practical cells (e.g. dry cell, lead acid battery, and fuel cell)

(3) Electrolysis

Electrode reaction, Faraday's law (molar change of substance and quantity of electricity), electrolytic refining

(4) Reaction rate and chemical equilibrium

Reaction rate and rate constant, reaction rate and concentration/temperature/catalyst, activation energy, reversible reaction, equilibrium constant, chemical equilibrium shift, Le Chatelier's principle

(5) Electrolytic dissociation equilibrium

Strength and degree of dissociation of acid and base, ionic product of water, dissociation equilibrium of weak acid and weak base, hydrolysis of salt, buffer solutions

III Inorganic Chemistry

1. Inorganic substances

(1) Typical elements (main group elements)

Properties, reactions, and uses of representative elements of each group and their compounds

Group 1: hydrogen, lithium, sodium, potassium

Group 2: magnesium, calcium, barium

Group 13: aluminum

Group 14: carbon, silicon, tin, lead

Group 15: nitrogen, phosphorus

Group 16: oxygen, sulfur

Group 17: fluorine, chlorine, bromine, iodine

Group 18: helium, neon, argon

(2) Transition elements

Properties, reactions, and uses of chromium, manganese, iron, copper, zinc, silver, mercury, and their compounds

(3) Industrial manufacturing methods of inorganic substances

(substances include sodium carbonate, aluminum, silicon, iron, copper, sodium hydroxide, ammonia, sulfuric acid, and nitric acid)

(4) Separation and analysis of metallic ions

(5) Widely used metals and inorganic compounds

Metals: titanium, tungsten, platinum, stainless steel, nichrome, etc.

Inorganic compounds: glass, fine ceramics, titanium oxide (photocatalyst), etc.

IV Organic Chemistry

1. Properties and reactions of organic compounds

(1) Aliphatic hydrocarbons

Structures, properties, and reactions of representative alkanes, alkenes, and alkynes; composition and uses of petroleum, etc.

Structural isomers and stereoisomers (cis-trans isomers, enantiomers)

(2) Aliphatic compounds with functional groups

Structures, properties, and reactions of representative compounds such as alcohols, ethers, aldehydes, ketones, carboxylic acids, esters, oils, soaps, etc.

(3) Aromatic compounds

Structures, properties, and reactions of representative compounds such as aromatic hydrocarbons, phenols, aromatic carboxylic acids, and aromatic amines

2. Organic compounds and our daily life

(1) Organic compounds essential to human functioning such as monosaccharides, disaccharides, and amino acids

(compounds include glucose, fructose, maltose, sucrose, glycine, and alanine)

(2) Main ingredients of typical drugs, dyes, and detergents

(ingredients include derivatives of salicylic acid, azo compounds, and sodium alkyl sulfate)

(3) Polymeric compounds

i) Synthetic polymers: structures, properties and syntheses of typical synthetic fibers and plastics

(compounds include nylon, polyethylene, polypropylene, poly(vinyl chloride), polystyrene, poly(ethylene terephthalate), phenol resin, and urea resin)

ii) Natural polymers: structures and properties of proteins, starch, cellulose, and natural rubber, and structures of nucleic acids such as DNA

iii) Applications of widely utilized polymers (e.g. water-absorbent polymers, conductive polymers, and synthetic rubbers); recycling of polymers as resources